LIGN 168 Computational Speech Processing - Final Project

Title: Classifying Speaker Gender Identity and Sexual Orientation using Wav2Vec2

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Supplementary Materials: Masterful. This is a link to the Google Colab notebook where you can run the analysis and generate the graphs. You'll need two types of files that are not housed in the notebook: 1) the labels.csv found here and 2) the .wav files also found here (there are 66 total). Drag and drop files 1) and 2) into the Files tab of the Colab notebook, and then run the cells sequentially. There is also a folder of all the figures found here so you can look at them more closely. A substantial portion of the loop for making the embeddings, classifier, and plotting was coded in tandem with ChatGPT-4o. A link to the conversation can be found here. The Wav2Vec2 speech recognition tutorial found here informed the embedding extraction as well.

Feedback: We would like detailed feedback, particularly with reference to possible next steps of things to try.

Self Assessment

Cover Page and Self-Assessment: Masterful. Cover page is included and all necessary elements are present.

Scope of Writeup: Masterful. Our writeup discusses all aspects of the project in detail with supplementary code and figures that demonstrate the specifics of how this experiment was done and what our results mean. The hypotheses are also explicitly addressed throughout the write-up.

Clarity of Methods: Acceptable. The code and data to reproduce the experiment is included in this write-up, and explanations of what was done and why it was done is included in the methods section. Admittedly, we could have explored other classification methods and our reasoning for choosing SVR linear regression is not fully fleshed out beyond starting simply with a linear classifier.

Demonstration of Knowledge: Masterful. We've discussed key concepts from neural ASR that are related to overall key concepts of the course such as framing and output of speech recognition systems as well as ethical considerations with interpretation.

Richness: Masterful. Our experiment engages multiple topics discussed in class including neural ASR, sampling and resampling, and ethical considerations of speech recognition systems. We discuss the experiment's pitfalls and strengths in detail, and provide nuanced discussion on the implications of these pitfalls and of our results in general. We also consider the data from different angles to ensure our interpretation didn't miss any low-hanging fruit.

Formatting and Length: Masterful. Our writeup is roughly 3000 words and was split evenly amongst us both. We think that the structure is clear for an implementation-type project, referencing the structure of an experimental paper, and that the length balances out well with the additional work done writing/editing code and preparing the results.

Structure and Organization: Masterful-to-Acceptable. Our paper is well-organized with necessary sections and subsections detailing our experiment, results, and discussion of said results. Our introduction fails to detail the structure and organization of the paper, but the overall structure is clear enough to make up for this and predictions are clear. There are some transitions between sections, but not for every section.

Language and Argumentation: Masterful-to-Acceptable. Ben initially wrote up the introduction and results while Henry initially wrote up the methods, discussion and conclusion. We then swapped and edited the opposite sections, checking for clarity and adding argumentation where necessary. While the clarity of the argumentation is there, we could probably have a little more engagement with the speech recognition/speaker identification literature to augment the demonstration of our knowledge learned from class.

Academic Integrity: Masterful. All references are included down below and any discussion or presentation of results are our own ideas and interpretations. Writing is our own and any code written with the help of generative AI is identified and cited as such in the supplementary materials.

Proposed Project Grade: 90%. We think this is a good grade to reflect the effort put into the analysis and the interpretation of our results. We put in a lot of effort to develop the story of the data and find the correct visualizations to tell that story as well as the correct methods for validating our interpretations. We think we could improve upon some of the argumentation and situating the paper beyond our course knowledge as well as a little tighter language and structure for telling that story.

<u>Write-up</u>

Introduction

Our experiment is guided by the question of whether or not an ASR model like Wav2Vec2 encodes speaker identity based on the feature vectors (embeddings) it generates from the acoustics of a speaker. Part of the main challenge in speaker identification tasks is "who said this utterance?" This question implies that any part of the speaker's identity that is available in the acoustic signal may influence the general identification of the speaker. This includes aspects of speaker identity such as gender identity and sexual orientation, which are part of a speaker's negotiation of their overall identity in speech (Gaudio, 1994; Linville, 1998; Pierrehumbert, 2004; Hazenburg, 2015; Zimman, 2017; Zimman, 2017; Houle and Levi, 2020). Our investigation below attempts to tease out the extent to which output embeddings of Wav2Vec2 contain information about speaker gender identity and sexual orientation by evaluating the predictions of a classifier trained on the output embeddings and comparing them to ground truth human perceptual ratings of speaker identity from Lang (2023).

Previous literature on human perception of gender identity and sexual orientation from speech has identified a swath of acoustic cues used for the perception of speaker identity: f0, formants, /s/, and creaky voice (Levon, 2006; Mack and Munson, 2012; Houle and Levi, 2021; Merrit and Bent, 2022; Lang

2023, Merrit et al. 2024). Any combination of these features, increasing or decreasing in the acoustic feature values in speech, has consequences for the perception of gender identity and sexual orientation in the mind of a human perceiver, making these acoustic features important landmarks for reconstructing the identity of a speaker from speech. Treating the neural ASR system of Wav2Vec2 as a comparable perceptual system, if Wav2Vec2 is sensitive to these same acoustic features for speaker identification, to the extent that they are included in the output embeddings, we can assess whether or not the information these acoustic features carry about identity is present in an embedding representation. This can be assessed via classification of speakers based on gender identity and sexual orientation using the extracted feature embeddings and comparing to a ground truth human perceptual rating of these aspects of speaker identity from prior experiments (Lang, 2023).

Wav2Vec2 is a pre-trained neural speech recognition model for extracting and identifying acoustic features in a signal (Baevski et al, 2020). Wav2Vec2 generates an embedding, or feature vector, from speech in an audio file, which can then be decoded to output individual speech sounds or transcripts of what was spoken. This embedding contains individual tensors, which are outputs of the final transformer layer in the neural network. Due to the black box nature of neural network models such as Wav2Vec2, attempting to solve the speaker identification problem with the output embeddings requires methods that probe or classify the output of the model. For example, de Seyssel et al. (2022) probed phoneme, language, and speaker representations in an unsupervised speech representation by extracting a vector representation of speech and comparing them in a two-dimensional space according to their individual phoneme, speaker, and language labels. Moreover, for speaker representation, the results indicated a binary split between speakers, for which gender was used as a proxy feature. Importantly, no claims were made about the model representations of gender, but this study shows the extent to which speaker identification representations remain relatively opaque as one piece of a many-faceted vector representation.

The question then remains of whether or not Wav2Vec2 encodes information about perceived speaker gender identity or sexual orientation in the extracted embeddings represented in the output of the neural ASR system, and whether or not this information is retrievable and useful for developing a model for speaker identification. If our classifier performs well (on either condition), then the embeddings in Wav2Vec potentially contain useful information regarding speaker identity. If our classifier performs poorly, the embeddings at minimum do not approximate human extraction of identity from the acoustic signal on this task, but could still be encoding speaker identity information.

Methods

Stimuli and Labels

The stimuli are taken from Lang (2023) and represent a collection of 66 WAV files of varying length. Each WAV file contains a short audio segment of speech (mean length = 1.5 seconds) from one speaker of American English, collected from podcasts. The labels for each of these 66 files are human ratings of perceived gender identity and perceived sexual orientation of the speaker on a standardized Likert scale from -1.5 to 1.5. For perceived gender identity, values closer to -1.5 indicate the perception of a more man-like individual while values closer to 0 indicate the perception of a non-binary or gender nonconforming individual, and values closer to 1.5 indicate the perception of a woman-like individual. For perceived sexual orientation, values closer to -1.5 indicate the perception of an individual that seeks relationships with a person of the opposite gender (e.g. heterosexual) while values closer to 0 indicate the perception of an individual that seeks relationships with a person of the any gender (e.g. bisexual, pansexual), and values closer to 1.5 indicate the perception of an individual that seeks relationships with a person of the same gender (e.g. homosexual).

Feature Extraction (Embeddings)

To create the feature embeddings of each of the 66 speakers from speech using Wav2Vec2, we used the WAV2VEC2_BASE model pre-trained on 960 hours of 16 kHz audio from the LibriSpeech dataset and not fine-tuned for any downstream task. As discussed in class, Wav2Vec2 works by generating latent representations every 10 ms. So, before extracting features, each waveform tensor shorter than the longest in the dataset was padded with silence using `torch.nn.functional.pad()` to ensure feature vectors of equal length. `torch.nn.functional.pad()` simply adds 0s to each waveform tensor before going through Wav2Vec2 to make sure each waveform tensor is equal length. Despite padding each waveform to result in equal length feature vectors, 6 of the resulting feature vectors were still about half as long as the rest and the missing values were filled in with NaNs. Because the classifier requires vectors of equivalent shape without NaNs, the samples that still had NaNs after padding were dropped for the sake of uniformity. We considered imputing the short feature vectors based on mean, median, or simply just replacing NaN with 0, but concluded that this introduced unnecessary complexity to the interpretability of the results. It is unclear what it means to impute or replace values in an embedding representation output from Wav2Vec2 so we concluded that adding "silence" (a string of 0s) to the end of the files to make them uniform in shape is the least opaque transformation of the data. Each input audio file was also resampled to 16 kHz per the specifications of Wav2Vec2.

After padding each waveform, we looped through `model.extract_features()` with each waveform to extract a feature vector for it, put that vector into a Numpy array, and then organized those arrays into a pandas DataFrame that contains each file name and that file's feature vector in the same order as the labels DataFrame. Labels here correspond to the average human perceptual rating for each speaker for the two conditions, perceived gender identity and perceived sexual orientation. The labels were then concatenated to the features DataFrame, and the redundant file names were dropped.

Classification

We used scikit-learn's Support Vector Regression linear classifier `SVR()` iterated 10,000 times on each speaker prediction with a Leave-One-Out method. Data preprocessing involved scaling the labels and embeddings to ensure that no unusually large values dominate the classifier and hurt its accuracy. Data from the 60 speakers and corresponding labels were then used to train and test the classifier. Classifiers were trained and tested separately on the labels from the two conditions, perceived gender identity and perceived sexual orientation. For every one speaker, the 59 other speakers were used to train the classifier via `LeaveOneOut()`, and then the classifier was tested on the embedding of the held out speaker and predicted the label. This process was iterated over for each speaker such that the predictions reported on in the results indicate scores for predicted values for each speaker based on a classifier trained on 59 other speakers' embeddings and labels.

Metrics

In order to establish a metric for performance of the classifier against the human perceptual ratings, Root Mean Squared Error (RMSE) was calculated using numpy `sqrt()` and sklearn mean_squared_error(). The RMSE was calculated between the true values (human perceptual ratings of gender identity and sexual orientation) and predicted values (classifier ratings of gender identity and sexual orientation). RMSE indicates the average variance of a given speaker from the best fitting classifier as represented by a linear regression through the true values and predicted values. Higher RMSE values indicate more variance and less accurate predictions while lower RMSE values (closer to 0) indicate less variance and better classification.

Results

The following section describes the results of classification for perceived gender identity and perceived sexual orientation based on Wav2Vec2 embeddings. In each graph, the true values, human perceptual ratings from Lang (2023), are compared to predicted values, classifier ratings of gender identity and sexual orientation. As described above, RMSE was calculated between the true values and predicted values within each condition. For gender identity, the RMSE calculated between true values (human perceptual ratings) and predicted values (classifier rating) is 0.89. For sexual orientation, the RMSE calculated between true values (human perceptual ratings) and predicted values (classifier rating) is 0.69. Figures 1, 2, 4, and 5 represent the relationship between the true values and predicted values for individual conditions. Figure 3 represents the relationship between the two conditions as described by the true values and predicted values. RMSE is most relevant for Figures 4 and 5.

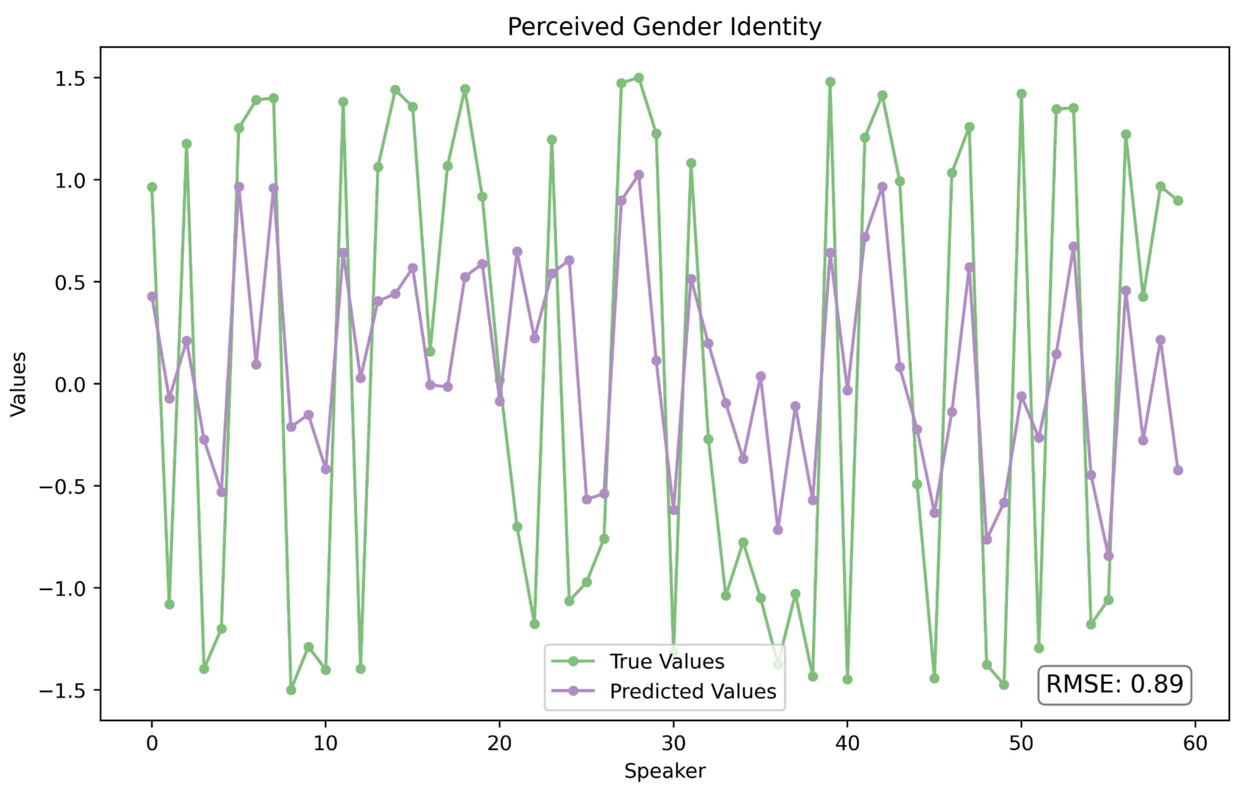

Figure 1. True values vs. predicted values for Perceived Gender Identity.

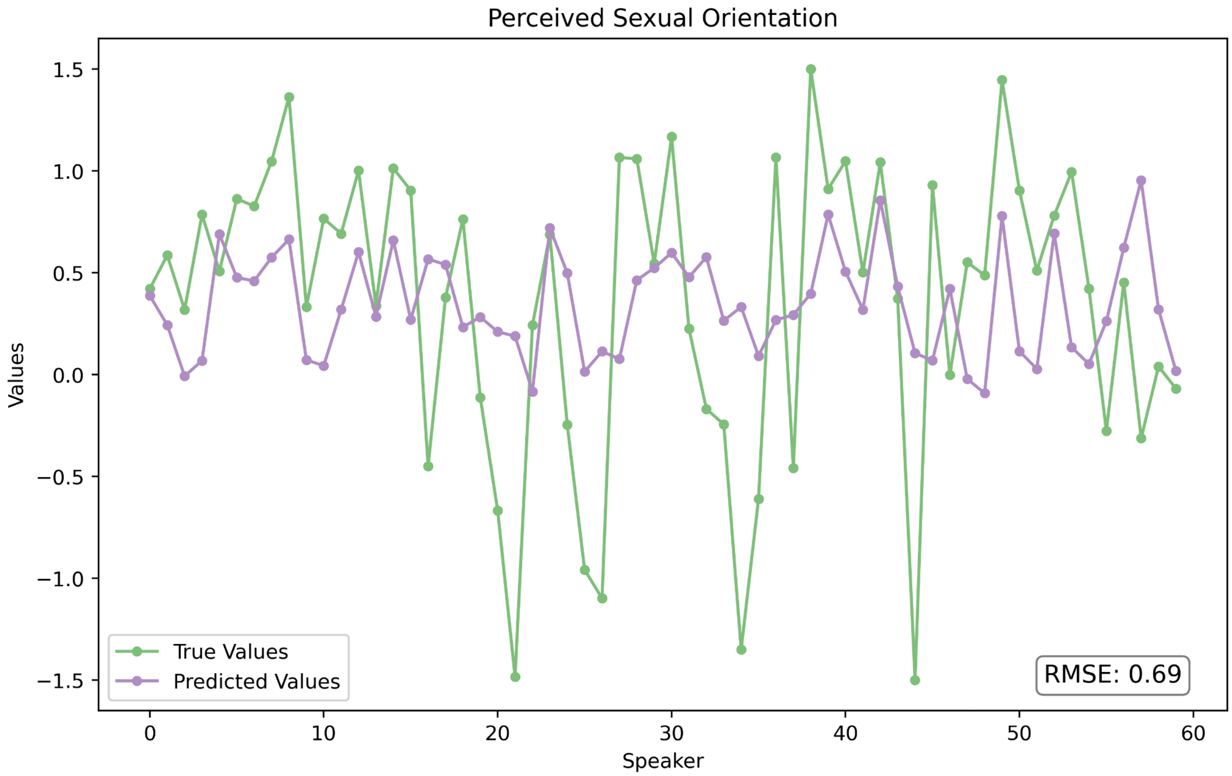

Figure 2. True values vs. predicted values for Perceived Sexual Orientation.

In Figures 1 and 2, each point on the x-axis represents a particular speaker from the stimulus set, and the two lines represent either the true values, meaning the average human perceptual ratings for a speaker, or the predicted values, the classifier ratings for each speaker. Figure 1 illustrates the ratings for gender identity for each speaker and Figure 2 illustrates the ratings for sexual orientation for each speaker.

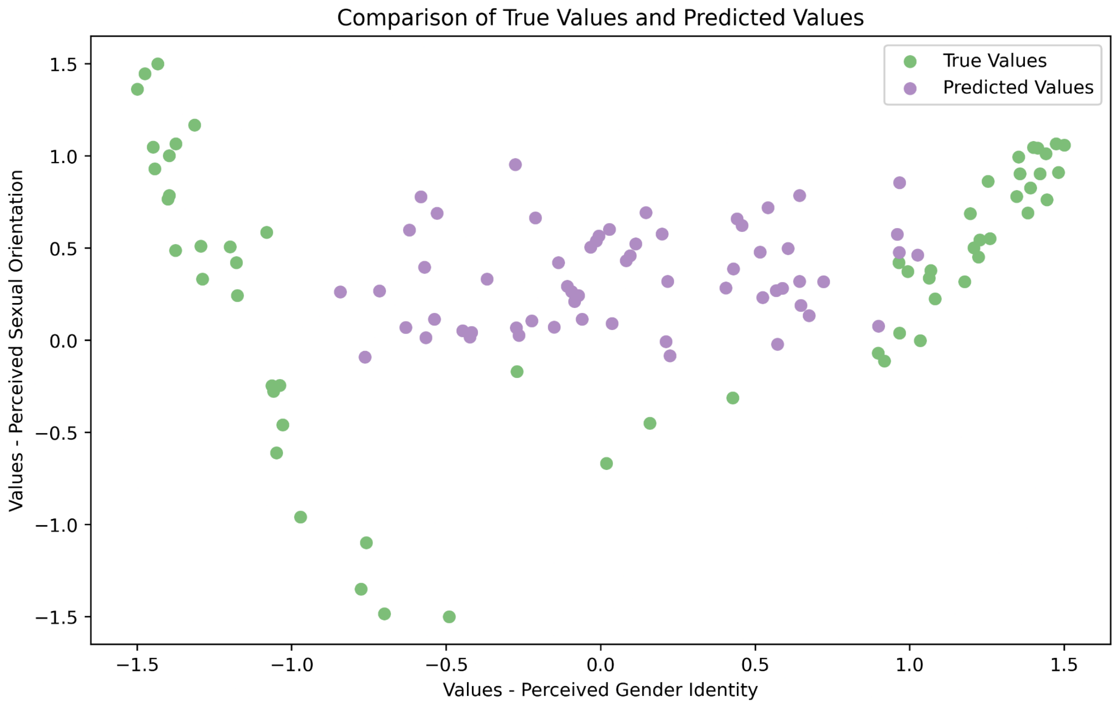

Figure 3. True values vs. predicted values for Perceived Gender Identity and Perceived Sexual Orientation.

In Figure 3, the points are individual speakers and their position along the ratings for the two conditions, gender identity and sexual orientation. The colors indicate the placement of a speaker within the two-dimensional space created by the gender identity and sexual orientation dimensions. In green, the true position of speakers as rated by humans is shown. In purple, the predicted position of speakers as rated by the classifier and informed by the embeddings is shown. Predicted values show a generally more dense cluster at the center of the two rating scales for gender identity and sexual orientation while true values show a stronger separation on gender identity and significantly more spread variation for sexual orientation.

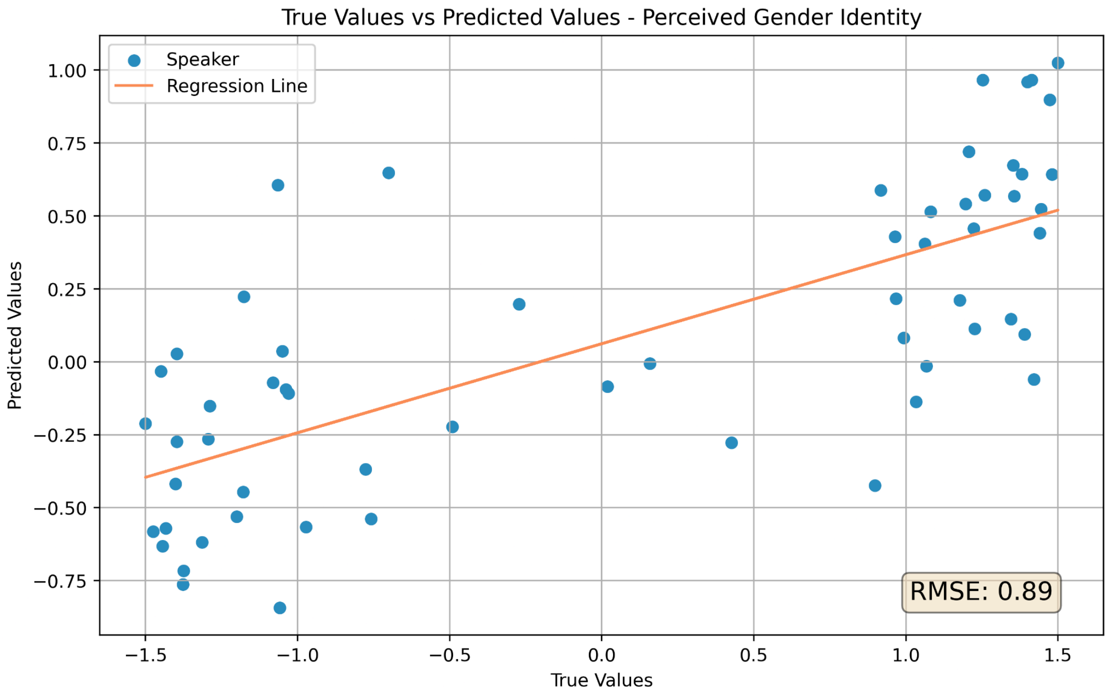

Figure 4. True values vs. predicted values for Perceived Gender Identity.

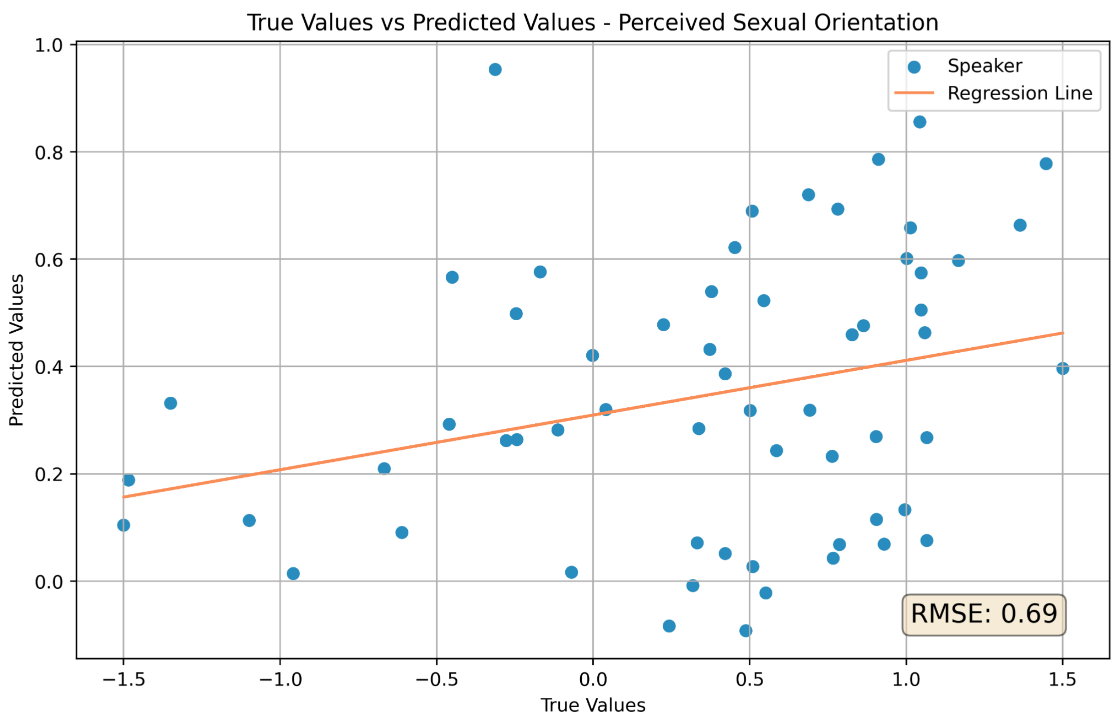

Figure 5. True values vs. predicted values for Perceived Sexual Orientation.

In Figures 4 and 5, each point represents a particular speaker's position in true values and predicted values. The line through the graph represents the points where the predicted values best match the true values. Lower RMSE indicates a better fit of the predicted values to the true values while higher RMSE represents a worse fit. With reference to the conditions, better fit indicates that the classifier is performing the perceptual rating task of speaker identity in a way that approximates the human perceptual rating. In Figure 4, the RMSE is higher relative to the RMSE in Figure 5. Visually speaking, the line of best fit through the gender identity predictions in Figure 4 fails to capture the variation along the scale, and in particular, the apparent clustering of two groups on either side of the true scale. In Figure 5, the line fits better, suggesting that the true values and predicted values have a more similar spread of variance around the best fit, but there's still generally a lot of variance.

Discussion

In Figures 1 and 2, the results indicate that the estimated predicted values for gender identity and sexual orientation do not easily approximate the true values. If the classifier was predicting either gender identity or sexual orientation labels with strong accuracy, we would expect the lines for predicted values in Figures 1 and 2 to approximate the lines for the true values, suggesting that both the training on the 59 other speakers was sufficient, and that the information in the embeddings was a useful representation of extracted acoustic features, including those features associated with the perception of a particular aspect of a speaker's identity. However, the lines in both Figure 1 and Figure 2 do not show a particularly strong relationship between predicted and true values. The same can be said for Figures 4 and 5 where the line of best fit does not do well at describing the variance in the data for both perceived gender identity and perceived sexual orientation.

The underfitting of the classifier based on the 60 feature vectors extracted from audio files could suggest that Wav2Vec2's feature vectors do not contain representations of speaker identity that correlate with gender identity and sexual orientation, though this underfitting could also be simply due to an insufficient amount of data for the classifier to train on. The tentative success due to generally lower and different scores of RMSE for both conditions of the classifier does suggest that it is learning something from the embeddings and using that to generate predictions, though without further investigation it is not clear if this is speaker identity. Even though the input was acoustic information, it is not clear that the acoustics represented within each feature vector are the same acoustic features that a human speaker might focus on when perceiving speaker identity. As discussed in class, Wav2Vec2 works by first creating latent representations of the waveform every 10ms in a CNN before combining them in the context network into context representations for use in transcription. Because we are using just the latent feature representations from the CNN, it is likely that the context network and its corresponding context representations would produce different results. Figure 3 best illustrates this interpretability problem because the clustering of all the predicted values near the center of the scales (values nearer to 0) does not mean that the predictions are just weaker and more conservative estimates away from chance. The center of the scale is itself a possible speaker identity, and it's unclear whether or not the model is learning to treat speakers that fall within that range as a kind of default position or if it is actually conservatively estimating the perceived gender identity and sexual orientation.

Methods for "looking into the black box" of neural ASR systems such as those discussed during the "Speaker and Language Classification" lecture and in de Seyssel et al. (2022) often employ smaller "probe" models that attempt to investigate what is stored within latent representations. Our experiment investigates Wav2Vec2's feature vectors in a similar way, and our results suggest that these vectors represent at least some acoustic information correlating with speaker identity broadly. However, whether or not specific representations of gender identity and sexual orientation are contained within a broad speaker identification representation are not supported by the evidence provided in our results. It is thus difficult to conclude that Wav2Vec2 feature vectors contain information about the speaker's identity. We can comfortably say that there is some kind of acoustic information contained within these representations, but because Wav2Vec2 is a system mostly meant for audio transcription, it could be possible that the model has simply not learned to strongly represent speaker identity.

Conclusion

Our experiment aimed at using a small linear regression classifier to determine whether feature embeddings made by Wav2Vec2 contain information about a speaker's gender identity and sexual orientation. We used a dataset of 66 short audio files with human ratings of perceived gender identity and sexual orientation from Lang (2022) which were all padded with 0s to ensure equal length embeddings. We generated predictions based on these embeddings by training scikit-learn's SVR linear classifier on 59 of the samples and testing on a single held out sample for every audio file in the dataset, and evaluated the performance of this classifier using Root Mean Squared Error (RMSE). For gender identity, the RMSE between predicted and actual was 0.89, and for sexual orientation it was 0.69. Generally, the classifier failed to capture the variance in the dataset, demonstrating slightly better performance on sexual orientation than on gender identity, but still overall subpar performance. These results suggest that Wav2Vec2's feature vectors do not contain strong enough representations of acoustic information that is associated with gender identity and sexual orientation, but it is also possible that the model is underfitting due to a lack of sufficient data.

Some possible next steps would be to try other methods of classification that more accurately capture the nonlinearity of the data like random forests or k-nearest neighbors, or even multi-step approaches that classify the feature vectors without the use of the labels in order to provide insights into the clusters of data we saw in the results of this experiment (see Figures 3 and 4). Another improvement would be to simply collect more data with a diverse selection of speakers. Ideally this data would consist of the same phrase read at about the same duration to isolate speaker specific information in the feature vector. This would likely ameliorate the severe underfitting of the classification model whilst combatting the need to pad samples with silence or impute feature vectors with made-up values. Overall, the underfitting and interpretability problem limited the scope of the results, but the results on their own appear to represent a moderately strong first step towards discovering the details of speaker identity in the broader speaker identification problem.

References

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